THIS/THAT

Redesigning a community poll app around one finding: people vote differently once they can see the crowd.



TOO CLOSE TO CALL · N = 1,204

ROLE
Research, product design, full-stack engineering

CONTEXT
CS7025, Trinity College Dublin. Node.js, Express, MySQL

AUTHOR
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From a working prototype to a product with a point of view

This or That lets people post a decision they are stuck on and have the community vote. The first version worked, but it looked generic, showed placeholder content, and had security problems serious enough to block a public launch. This study covers the audit, the research behind each design decision, the new visual system and the plan for the backend.

9
problem areas found in the audit, 4 of them critical

6
peer-reviewed or primary sources shaped the design

7
screens designed, desktop and mobile

The three decisions that define the redesign

- **Vote, then reveal.** Results stay hidden until you vote. Three experiments show that visible tallies pull later votes toward the leader.
- **Strictly two sides.** One question, two options, shown as a single split bar with a 50% line. Every result reads at a glance.
- **A results-desk vocabulary.** Polls are called like elections: too early, too close, leaning, landslide. Each label maps to a rule on the server.

What changed for users

- Voting is one tap with no page reload, and the split animates in.
- Arguments are grouped by side, so you read the strongest case for each option.
- Askers report what they chose in the end, which closes the loop for voters.
- A profile shows your record: how often you side with the crowd, by topic.
- A live check helps people write better questions before posting.

What was there

I read every file in the codebase: 6 controllers, 6 route files, 7 pages, a 2,223-line stylesheet, the schema and the seed scripts. The backend structure was sound. The problems were in configuration, data integrity and the front end.

CRITICAL	**Production database password committed to the repo.** It was base64-encoded as a fallback in `config/db.js`, beside a hard-coded JWT secret. Anyone with the repo could read the database or forge a login. OWASP A02 Security Misconfiguration.
CRITICAL	**Stored cross-site scripting.** Usernames and avatar data reached `innerHTML` and an `<img src>` attribute without escaping. OWASP A05 Injection.
CRITICAL	**Schema out of sync with the code.** `schema.sql` lacked seven columns the code reads, including email and display name, so a fresh install failed at sign-up.
CRITICAL	**Topics that silently did nothing.** The interface offered 16 topics, but the database held 6 differently named ones. Most picks were dropped without an error, so personalisation rarely worked.
HIGH	**Votes and comments shared one row.** A unique key on (question, user) meant each person could leave only one comment, and a new one overwrote the old. Votes also weren't checked against the question, so an option from another poll was accepted. OWASP A01 Broken Access Control.
HIGH	**Fake content in the live UI.** The hero (“1,240 active votes”), leaderboard, trending topics and profile defaults (“Kelly”, ID TT-2025-00421) were all hard-coded.
HIGH	**Feeds that didn't do what they said.** “New” called the same endpoint as “For you”. “Trending” sorted by all-time totals. “Unanswered” sorted by fewest votes, so answered polls still appeared.
MEDIUM	**No hardening.** No rate limiting on login, no security headers (although the README said there were), a 10 MB request limit, and a full page reload after every vote.
MEDIUM	**Dead controls.** Save buttons, the question “type” field and the comments-off setting were stored or drawn but never acted on.

Showing the score changes the score

The old app showed live percentages before you voted. The research on social influence says that design is the main threat to a poll's usefulness, since a poll is only worth anything if the votes are independent.

SALGANIK, DODDS & WATTS · SCIENCE · 2006

MusicLab: social influence makes outcomes unequal and unpredictable

Participants were randomly assigned either to an independent condition or to one of eight “worlds” where they could see download counts. In the social worlds, success was more unequal (higher Gini coefficient) and harder to predict. The best songs rarely did badly and the worst rarely did well, but almost anything else could happen.

→ DESIGN DECISION

Hide the tally until the person has voted. Each vote then reflects the voter's own view, not the counter's.

MUCHNIK, ARAL & TAYLOR · SCIENCE · 2013

One early upvote measurably inflates the final score

In a five-month randomised experiment on a Reddit-like news site, one artificial upvote raised the chance of the next rating being positive by about 32%, and final ratings ended about 25% higher on average. The effect was asymmetric: people corrected negative manipulation but piled onto positive.

→ DESIGN DECISION

The feed doesn't rank by raw vote count. It shows no counts before you vote, only the number of votes so far, never who is ahead.

LORENZ, RAUHUT, SCHWEITZER & HELBING · PNAS · 2011

Seeing others' answers narrows a crowd without making it more accurate

144 participants made factual estimates, with money at stake for accuracy, and could revise them after seeing others' answers. Information sharing reduced the diversity of answers without improving the group average, while making people more confident.

→ DESIGN DECISION

Arguments are hidden until you vote too, and they're grouped by side, so a reader meets the best case for each option instead of a pile-on.

Trade-off. Hiding results makes browsing less satisfying for someone who only wants to see outcomes. The poll page will offer “Skip and see results”, which records a non-vote, so lurkers are still served and the independence of the data is kept.

Why two options, and how to rank them

SCHEIBEHENNE, GREIFENEDER & TODD · J. CONSUMER RESEARCH · 2010

Choice overload is weaker than its reputation

A meta-analysis of 63 conditions from 50 experiments (5,036 participants) found a mean choice-overload effect of virtually zero, with large variation between studies. More options tended to help when people already had clear preferences.

→ WHAT I DIDN'T CLAIM

“Fewer choices are always better” isn't supported by the evidence, so it isn't the reason for a binary format. The reason is legibility: with two options, one bar, one margin and one verdict say everything.

HACKER NEWS RANKING · news.arc, ANALYSED BY K. SHIRRIFF

Time decay keeps a feed fresh

The commonly cited form is `score = (votes − 1) / (hours + 2)^1.8`. The exponent, called gravity, controls how fast older items sink. The live site also applies penalty factors, for example to posts with many comments relative to votes.

→ DESIGN DECISION

“Rising” uses votes and arguments from the last 48 hours, divided by `(age + 2)^1.5`. Polls that are close get a small boost, since a near-even split is the most useful one to add a vote to.

E. MILLER · “HOW NOT TO SORT BY AVERAGE RATING” · 2009

Small samples lie

Sorting by a raw percentage ranks 2 out of 2 above 100 out of 101. Miller recommends the lower bound of the Wilson score confidence interval, which takes sample size into account.

→ DESIGN DECISION

Below 10 votes a poll is labelled “Too early to call” and gets no “leaning” label. Argument ranking uses the Wilson lower bound of the share of readers who marked it convincing.

The verdict scale

TOO EARLY TO CALL — n < 10

TOO CLOSE TO CALL — margin ≤ 6 pts

LEANING THIS — leader below 70%

LANDSLIDE · THAT — leader at 70% or more

What exists, and what to avoid looking like

PRECEDENT · GAS (2022)

Curated polls can spread fast, but at a cost

Gas reached #1 on the US App Store within about three months of launch, and Discord acquired it in January 2023. Its polls were written by the company, not by users, and growth relied on referrals that schools found intrusive. Much of the usage data is the founder's own.

Lesson: well-written prompts drive engagement. The redesign seeds every topic with carefully written questions but keeps user-written ones as the core.

PRECEDENT · INSTAGRAM POLL STICKERS

One tap is the lowest-effort form of participation

Instagram has reported that about 60% of businesses posting Stories add an interactive element such as a poll each month. Published “engagement rate” benchmarks for polls mostly come from marketing blogs without sources, so I didn't rely on them.

Lesson: voting must take one tap and give instant feedback. Commenting stays optional and comes after the vote.

VISUAL RESEARCH · THE “AI-MADE” LOOK

The design defaults that make an app look generated

Design commentary in 2025-26 describes a cluster of tells: a purple or indigo gradient hero, the Inter typeface as the safe pick, a centred hero followed by three identical rounded cards with icons, and gradients used as background decoration. Several writers trace the purple to Tailwind's default palette, which dominated the tutorials models learned from.

Instead of a gradient
Two flat inks with fixed meanings: denim always means THIS, marigold always means THAT.

Instead of Inter
A condensed scoreboard face for numbers, a newsroom grotesk for text, a mono for counts.

Instead of feature cards
The product opens on the real feed, with a poll you can vote on in the first screen.

Design principles from the research

1. Independence first. Nothing on screen tells you how others voted until you've voted.

2. Honest numbers. Always show n. Never call a result on a small sample.

3. One tap to take part. Voting never needs a page load, a confirmation or an account prompt mid-flow.

4. Close the loop. Askers report their final choice, so voters learn whether their advice was taken.

Two Inks

Every decision has two sides, so the interface has two inks. Denim and marigold are used only to mean THIS and THAT. Everything else is ink on a cool paper grey, so the split is always the loudest thing on the screen.

THIS	THAT	Aa	Aa	Aa	Aa
Denim	**Marigold**	**Ink**	**Slate**	**Paper**	**Rule**
`#4A6B9C · 5.4:1`	`#FFC21A · 11.1:1 ink`	`#14152B`	`#3D4160 · 9.9:1`	`#EEF0F4`	`#DADCE6`

Contrast ratios are WCAG 2.x, measured against white unless noted. The first draft used a saturated cobalt. Review feedback found it too heavy, so it was replaced with a calmer denim that still passes AA for body text.

79%

Big Shoulders Display. A condensed face with scoreboard roots, used for percentages, verdicts and headings, never for sentences.

Lisbon or Porto for a first long weekend?

Schibsted Grotesk. Designed for a news publisher, with sturdy figures. Used for questions, body copy and controls.

`n = 295 · margin 58 pts`

IBM Plex Mono. Used for counts, timestamps and anything a reader might compare down a column.

The signature component: the split



One bar, two inks, and a dashed line at 50%, borrowed from election-night graphics. Before you vote, it appears as two tappable halves with an “OR” between them. After you vote, the boundary slides to the result.

Seven screens, one flow

SCREEN	WHAT IT DOES	GROUNDED IN
Feed	Lenses (For you, Rising, Fresh, Needs votes, Saved). Cards show two tappable options and only the vote count. After you vote, the split, verdict, n and whether you sided with the majority. Undo, save and copy link.	Vote-then-reveal (Salganik; Muchnik); time decay (HN)
Poll detail	A results desk with large percentages, a verdict and margin, and a chart of how the split moved over time. Arguments are split into “For THIS” and “For THAT” columns and sorted by how convincing readers found them.	Lorenz et al.; Wilson lower bound (Miller)
Ask	Composer with a live preview card and a question check: is it a question, are the options distinct, do they fit on a button, is there context. Topic, closing time, anonymity, arguments on or off, hidden results.	Quality of prompts (Gas precedent)
Profile	Your decision record: votes cast, polls asked, the share of your votes that sided with the majority by topic, and polls waiting for you to report an outcome.	Closing the loop
Mobile feed	One decision per screen. The screen is split into two tap targets. After a tap, the halves resize to the result and “Next” appears.	One-tap participation
Onboarding	Pick three or more topics. Each tile shows an example question, so the choice is concrete.	Fixes the topic mismatch in the audit
System sheet	Tokens, type, verdict scale, components and dark-mode values.	—

New features

- Outcome reports from askers, with notifications to voters
- Arguments grouped by side, with “convincing” marks
- Closing times and automatic closing
- Save, search and topic filters
- Crowd-alignment stats by topic
- Edit context before the first vote; delete your own poll
- Report a poll or argument; asker can turn arguments off

Accessibility as drawn

- Real buttons, links and labelled inputs throughout
- Touch targets of at least 44 px
- THIS and THAT differ in lightness as well as hue, so they stay distinct for colour-blind users and in greyscale
- Visible focus states; motion respects reduced-motion settings
- Every percentage is printed as text, never only as bar length

Hardening the engine

The redesign needs a backend that enforces its rules. Each change below answers an audit finding or a category in the OWASP Top 10 (2025). This section is the implementation plan; the build follows sign-off on the design.

CHANGE	DETAIL	ANSWERS
Secrets out of code	All credentials come from environment variables. The server refuses to start without them. Ship `.env.example`. Rotate the leaked database password.	A02 Misconfiguration
Separate votes from arguments	`votes` (one per user per poll, enforced by a key) and `arguments` (many per user, each tagged with a side). A vote is only accepted if the option belongs to the poll and the poll is open.	A01 Access Control
Validation at the edge	Every request body is checked against a schema (types, lengths, enums). Text is stored raw and escaped on output; the front end renders with `textContent`.	A05 Injection
Headers and limits	Helmet with a strict Content Security Policy. Rate limits on sign-in, sign-up, voting and posting. 100 KB request limit. Avatars uploaded as files and type-checked.	A02, A07 Authentication
Ownership checks	Edit and delete are allowed only for the asker, and context edits only before the first vote. Anonymous posts never expose the author ID in an API response.	A01 Access Control
Fail closed	One error handler with consistent JSON errors and no stack traces in responses. Process-level handlers log and exit cleanly instead of carrying on in an unknown state.	A10 Exceptional Conditions
Fast, correct feeds	Vote counts are stored per option and updated in the same transaction as the vote. Feeds use cursor pagination. Rising, Fresh and Needs-votes are real queries.	Audit: feeds
Migrations and tests	Versioned migrations that keep existing data. Integration tests for auth, voting rules and ownership against a disposable MySQL. Dependency audit in CI.	A03 Supply Chain

A question set people recognise

The old seed data used six generic categories that didn't match the interface. The new set has twelve topics, each seeded with questions that are specific, local to the Dublin student audience and genuinely two-sided.

TOPIC	EXAMPLE QUESTION	THIS	THAT
TECH	MacBook Air or ThinkPad X1 Carbon for a design + dev master's?	MacBook Air	ThinkPad
MONEY	Clear a €4k credit union loan first, or start investing €200 a month?	Clear the loan	Start investing
CAREER	Fully remote startup, or a hybrid big-firm role paying €6k more?	Remote startup	Hybrid big firm
STUDY	Berkeley Library or the Hamilton for exam season?	Berkeley	Hamilton
TRAVEL	Lisbon or Porto for a first long weekend in Portugal?	Lisbon	Porto
FOOD & DRINK	Meal-prep on Sunday, or cook fresh every night?	Meal-prep	Cook fresh
HOME	Rent alone in Rathmines, or share in Phibsborough and save €450?	Live alone	House-share
HEALTH	Gym at 7am or 7pm when you're new to it?	7am	7pm
STYLE	One good wool coat, or two cheaper jackets for a Dublin winter?	One good coat	Two jackets
RELATIONSHIPS	Tell a friend you can't afford the group holiday, or quietly find the money?	Tell them	Find the money
ENTERTAINMENT	Read Normal People first, or watch the series first?	Book first	Series first
EVERYDAY	Luas or cycle to campus through a Dublin winter?	Luas	Cycle

Writing rules for seed questions: name the real trade-off, include one concrete constraint (budget, time, place), keep each option under 28 characters, and avoid questions with an obvious right answer.

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A note on evidence. Sources 1–6 are peer-reviewed papers or analyses of public source code. Sources 8–10 are journalism and practitioner commentary, used only for market context and visual direction. Where a figure came only from a company or an unsourced blog, the study says so or leaves it out.